SafeEd - Secure University Record System

Semester Project Report

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1. Abstract

SafeEd is a secure university record system which is developed as a group project. The main purpose of this project is to protect student academic information and control access of users. This is a console based system developed in Python language. In this system teachers and students login using username and password. Passwords are not stored in simple form, they are converted into hashed form using SHA-256 algorithm. Students are only allowed to view their own records while teachers can view and update records of assigned students only. The system also maintain a security log file to record login, logout and update activities. This project give basic idea of secure record management system.

2. Introduction

In today universities, student data security is very important issue. Many systems are not properly secure and student data can be misuse easily. Unauthorized access and data modification is common problem in weak systems. SafeEd project is designed by us to reduce these issues. This system apply basic security rules to protect academic records. Only authorized users are allowed to access specific information. The project is implemented using Python programming language and run in console environment.

2.1 Objectives

The main objective of this project is to develop a secure system for managing university student records. Another objective is to understand and implement authentication and authorization concepts practically. This project also aim to apply password hashing technique for improving security. One more objective is to maintain system activity logs for monitoring and security purpose.

2.2 Problem Statement

Universities face many problems related to data security and access control. Student records can be accessed or modified by unauthorized persons. In many systems there is no proper role based access mechanism. Manual record keeping is also not secure and chances of errors are high. Therefore, there is a need of a secure system which can control access and protect student academic data.

2.3 Project Scope

The scope of this project is limited to a console based application. It includes login system for teachers and students. Students can view only their grades, attendance and fee status. Teachers can view and update records of only assigned students. A security log file is used to store important system events. Database system and graphical user interface are not included in this project.

3. Relevant Background and Definitions

Authentication is the process of verifying the identity of a user by using username and password. Authorization means giving permission to users to access specific data. Hashing is a technique used to convert passwords into unreadable form. SHA-256 is a secure hashing algorithm which is used in this project. Confidentiality means that data is only accessible by authorized users. Integrity ensures that data is not changed without proper permission.

4. Literature Review and Related Work

Many student record management systems have been developed in recent years. Most of these systems use role based access control to secure student data. Research shows that storing passwords in hashed form is important to prevent data breaches. Security logging is also used in many systems to track user activities. SafeEd project follows these basic concepts on a small scale for academic and learning purpose.

5. Experimental Evaluation and Results

The SafeEd system was tested using multiple teacher and student accounts. The login system successfully allow access to valid users and reject invalid login attempts. Students were able to view only their own records and not others data. Teachers were able to view and update only the records of assigned students. Unauthorized access and update attempts were blocked by the system. All activities such as login, logout and record updates were saved in the security log file. The results show that the system work according to defined security requirements.

6. Conclusion

In conclusion, SafeEd is a secure university record system developed as a group project. The project successfully implement basic security concepts such as authentication, authorization and activity logging. The system maintains confidentiality and integrity of student academic records. Although the system is simple and console based, it provides clear understanding of secure system design. In future, this project can be enhanced by adding database support, graphical interface and more advanced security features.